**Note**

Viewing in the positive direction of $y-$axis, the curve $y=f(x)$ can be termed as the upper curve (U) and the curve $y=g(x)$ as the lower curve (L). Thus, we get $A=\int_a^b [y_U - y_L]dx$.

**Case (ii)**

Let $x=f(y)$ and $x=g(y)$ be the equations of two curves in the $XOY-$plane such that $f(y)\geq g(y)$ for all $y\in[c,d]$. We want to find the area $A$ of the region bounded between the two curves, the lines $y=c$ and $y=d$. The required area is sketched in Fig. 9.24. To compute $A$, we view in the positive direction of the $x-$axis and divide the region into thin horizontal strips of width $\Delta y$ and height $f(y)-g(y)$. It is important note that $f(y)-g(y)\geq 0$ for all $y\in[c,d]$. As before, the required area is the limit of the sum of the areas of the horizontal strips. Hence, we get $A=\int_c^d [f(y)-g(y)]dy$.

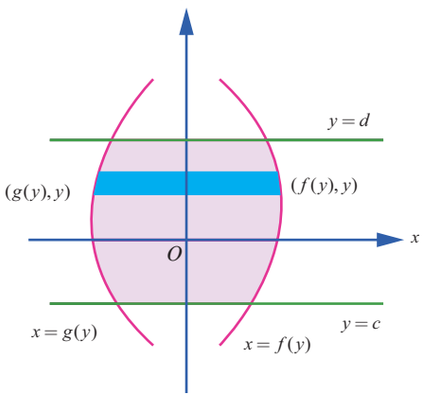


Fig. 9.24

**Note**

Viewing in the positive direction of $x-$axis, the curve $x=f(y)$ can be termed as the right curve (R) and the curve $x=f(y)$ as the left curve (L). Thus, we get $A=\int_a^b [x_R - x_L]dy$.

**Example 9.54**

Find the area of the region bounded between the parabolas $y^2=4x$ and $x^2=4y$.

**Solution**

First, we get the points of intersection of the parabolas. For this, we solve $y^2=4x$ and $x^2=4y$ simultaneously: Eliminating $y$ between them, we get $x^4=64x$ and so $x=0$ and $x=4$. Then the points of intersection are $(0,0)$ and $(4,4)$. The required region is sketched in Fig.9.25.

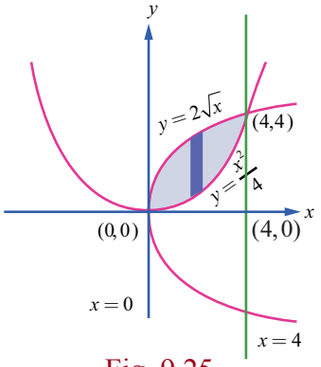


Fig. 9.25

Viewing in the direction of $y$-axis, the equation of the upper boundary is $y=2\sqrt{x}$ for $0\leq x\leq 4$ and the equation of the lower boundary is $y=\frac{x^2}{4}$ for $0\leq x\leq 4$. So, the required area $\Delta$ is

$$A=\int_0^4 (y_U - Y_L)\,dx=\int_0^4\left(2\sqrt{x}-\frac{x^2}{4}\right)dx=\left[2\left(\frac{2x^{3/2}}{3}\right)-\frac{x^3}{12}\right]_0^4=\left[2\left(\frac{2\times 8}{3}\right)-\frac{64}{12}\right]-0=\frac{16}{3}.$$

**Note**

Viewing in the positive direction of $x$-axis, the right bounding curve is $x^2=4y$ and the left bounding curve is $y^2=4x$. See Fig. 9.26. The equation of the right boundary is $x=2\sqrt{y}$ for $0\leq y\leq 4$ and the equation of the left boundary is $x=\frac{y^2}{4}$ for $0\leq y\leq 4$. So, the required area $A$ is

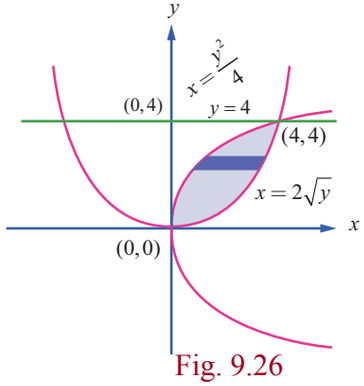


Fig. 9.26



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